

Luxury Real Estate Agent Portfolio — Flutter + Firebase

****Course project:**** A high-end mobile portfolio for a luxury real estate agent. The app uses a ****Dark Luxury**** UI (charcoal + gold), personal branding over raw listings, and ****Cloud Firestore**** to store and stream portfolio content.

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1. Overview

| Topic | Detail |

|-----|-----|

| ****Framework**** | Flutter 3.x / Dart 3.11+ |

| ****Backend**** | Firebase — ****Cloud Firestore**** for portfolio JSON |

| ****Images**** | Unsplash URLs (HTTPS) stored in Firestore / local seed |

| ****Theme**** | Dark Luxury — charcoal `#0B0C0E`, gold `#D4AF37`, ivory text |

| ****Platforms**** | iOS, Android, macOS, Web, Windows (Firebase via FlutterFire) |

****Design goal:**** Present the ****agent as the brand**** (mission, stats, awards, expertise) and use properties as curated highlights—not a full MLS feed.

2. Exact project structure

Challenge2/

- |— README.md ← This file (setup + structure + screenshots)
- |— pubspec.yaml ← Dependencies: firebase_core, cloud_firestore
- |— analysis_options.yaml
- |— firebase.json ← Firebase CLI — points to firestore.rules
- |— firestore.rules ← Firestore security rules (demo: open read/write)
- |
- |— docs/
 - | |— screenshots/ ← YOU add PNG/JPG captures here (see §7)
 - | | |— README.md ← Suggested screenshot filenames
 - | | |— .gitkeep
- |
- |— lib/
 - | |— main.dart ← Firebase init + MaterialApp + shell
 - | |— firebase_options.dart ← Platform Firebase keys (from FlutterFire CLI)
 - | |
 - | |— data/
 - | | |— local_portfolio_seed.dart ← Default content + Firestore seed source
 - | |
 - | |— models/
 - | | |— agent_models.dart ← PropertyListing, AgentStat, awards, etc.
 - | | |— portfolio_bundle.dart ← Aggregate model + Firestore map conversion
 - | |
 - | |— services/
 - | | |— portfolio_repository.dart ← Firestore read/write/stream + fallback
 - | |
 - | |— theme/
 - | | |— luxury_dark_theme.dart ← Material 3 dark luxury theme
 - | |
 - | |— shell/
 - | | |— agent_portfolio_shell.dart ← 4 tabs, StreamBuilder, seed banner
 - | |
 - | |— screens/

```

| | |—— agent_home_screen.dart ← Brand: SliverAppBar, mission, carousel
| | |—— stats_dashboard_screen.dart
| | |—— property_showcase_screen.dart
| | |—— property_detail_screen.dart ← Hero destination
| | |—— expertise_screen.dart ← Edge: awards + expertise
| |
| |—— widgets/
| | |—— glass_surface.dart ← BackdropFilter glass panels
| | |—— glass_bottom_navigation.dart
| | |—— luxury_network_image.dart
| | |—— luxury_property_card.dart ← Hero + status badge + glass overlay
| | |—— featured_success_carousel.dart
| | |—— estate_awards_carousel.dart ← Vertical awards list
|
|—— test/
| |—— local_portfolio_seed_test.dart
| |—— widget_test.dart
|
|—— android/ ← Flutter Android host + google-services.json (after configure)
|—— ios/ ← GoogleService-Info.plist (after configure)
|—— macos/
|—— web/
|—— windows/
...

```

****Where logic lives****

```

| Concern | Location |
|-----|-----|
| UI layout | `lib/screens/`, `lib/widgets/` |
| Navigation / tabs | `lib/shell/agent_portfolio_shell.dart` |
| Firebase I/O | `lib/services/portfolio_repository.dart` |
| Content defaults | `lib/data/local_portfolio_seed.dart` |
| Types + Firestore parsing | `lib/models/` |

```

3. Firebase data model

All portfolio fields are stored in ****one Firestore document**** for simple setup and marking:

Firestore path	Description
----- -----	
Collection `portfolio`	Root collection
Document `main`	Single agent portfolio record

****Document fields (map):****

Field	Type	Example
----- ----- -----		
`agentName`	string	`Jordan Mercer`
`agentTitle`	string	`Luxury Real Estate Advisor`
`missionStatement`	string	Long text
`profileHerolImageUrl`	string	Unsplash HTTPS URL
`stats`	array of maps	`{ label, value, subtitle? }`
`featuredSuccesses`	array	`{ headline, detail, imageUrl }`
`listings`	array	`{ id, title, imageUrl, priceDisplay, status, neighborhood, detailLine }`
`expertisItems`	array	`{ title, chips: string[] }`
`awards`	array	`{ title, issuer, year, tagline }`
`updatedAt`	string (ISO8601)	Set on seed

****Listing `status` values:**** `sold` | `exclusive` | `active`

The app listens with `snapshots()` so UI updates when you edit data in the Firebase Console.

****Offline / no Firebase:**** If Firebase is not configured, the app uses `LocalPortfolioSeed.bundle` automatically (useful for `flutter test` and first clone).

4. Configure Firebase (required)

4.1 Prerequisites

- [Flutter SDK](https://docs.flutter.dev/get-started/install)
- [Firebase account](https://console.firebase.google.com/)
- Node.js (for Firebase CLI, optional but recommended)

4.2 Create a Firebase project

1. Open [Firebase Console](https://console.firebase.google.com/) → **Add project** (e.g. `luxury-agent-portfolio`).
2. Enable **Cloud Firestore** → **Create database** → start in **test mode** (or production + deploy `firestore.rules` from this repo).
3. Register apps for the platforms you will run:
 - **Android** — package name: `com.example.challenge2` (see `android/app/build.gradle.kts`)
 - **iOS / macOS** — bundle ID: `com.example.challenge2`
 - **Web** — if using Chrome

4.3 Install FlutterFire CLI and link the project

From the project root:

```
```bash
cd /Users/roc/Desktop/MP/Challenge2

One-time: install FlutterFire CLI
dart pub global activate flutterfire_cli

Ensure pub cache is on PATH (add to ~/.zshrc if needed):
export PATH="$PATH:$HOME/.pub-cache/bin"

flutterfire configure
```
```

Follow the prompts:

- Select your Firebase project
- Select platforms (android, ios, macos, web, etc.)

This command:

- Generates / overwrites `lib/firebase\_options.dart` with real API keys
- Adds `google-services.json` (Android) and `GoogleService-Info.plist` (iOS/macOS)

****Verify:**** Open `lib/firebase\_options.dart` — `projectId` must ****not**** contain `REPLACE\_ME`.

`DefaultFirebaseOptions.isConfigured` will then return `true` at runtime.

4.4 Deploy Firestore rules (optional)

```
```bash
npm install -g firebase-tools
firebase login
firebase deploy --only firestore:rules
```
```

Demo rules in `firestore.rules` allow public read/write on `portfolio/\*\*` for coursework—****restrict before any real deployment****.

4.5 Install Flutter packages

```
```bash
flutter pub get
```
```

5. Run the application

```
```bash
cd /Users/roc/Desktop/MP/Challenge2
flutter pub get
flutter run
```
```

Examples:

```
```bash
flutter run -d macos
```

```
flutter run -d chrome
```

```
flutter devices # list emulators / devices
```

```
...
```

**\*\*macOS images:\*\*** Outbound network is enabled in macOS entitlements for Unsplash URLs.

**\*\*First launch without seed:\*\*** UI shows local demo data. After Firebase is configured, use **\*\*Seed\*\*** (see below) to persist data to Firestore.

```

```

## ## 6. Seed Firestore with demo data

### ### Option A — In-app (recommended)

1. Configure Firebase (§4) and run the app.
2. If Firestore document `portfolio/main` is missing, a banner appears: **\*\*"Firestore is empty — load demo portfolio"\*\*.**
3. Tap **\*\*Seed\*\*** → writes `LocalPortfolioSeed` to Firestore.
4. UI refreshes from the live stream (`sourceLabel: firebase`).

### ### Option B — Firebase Console (manual)

1. Firestore → **\*\*Start collection\*\*** → ID: `portfolio`
2. Add document ID: `main`
3. Copy field structure from `lib/data/local\_portfolio\_seed.dart` / `PortfolioBundle.toFirestoreMap()` in code, or run the seed once via Option A and inspect the document.

### ### Option C — Verify in console

After seeding, you should see collection `portfolio`, document `main`, with arrays `stats`, `listings`, `awards`, etc.

```

```

## ## 7. Screenshots

Jordan Mercer

Mission statement

I represent discerning buyers and sellers who expect discretion, data-backed guidance, and a calm hand through high-stakes transactions. Your move is personal – your brand experience should be too.

Featured successes



**Waterfront closing – 12% above ask**  
Private tour funnel + pricing discipline



**Estate sale in 9 days**  
Staging narrative + buyer matching



**Off-market tower duplex**  
Quiet matchmaking for principals



**Investor repositioning**  
Renovation ROI roadmap

  
Brand

  
Stats

  
Homes

  
Edge

Performance snapshot

Numbers support the narrative – they never replace it.

**YEARS EXPERIENCE**

**14**

Across luxury corridors

**HOMES SOLD**

**186**

Off-market + public

**AVG. DAYS ON MARKET**

**18**

Portfolio average

  
Brand

  
Stats

  
Homes

  
Edge



# Signature properties

Curated highlights — each card uses a shared Hero tag into the detail view.



EXCLUSIVE

\$4,850,000

Skyline Penthouse · Yorkville

3 bed · 3.5 bath · 3,200 sq ft · 360° views



SOLD

\$2,195,000

## Recognition & depth

Credentials open doors; judgement keeps them open.

### Recognition



2024

**Chairman's Circle — Diamond**

Greater Toronto Luxury Guild

Top 1% closed volume, ultra-luxury corridor



2023

**North Star Service Excellence**

Canadian Estate Council

Principal satisfaction & confidential placement



2022

**Luminary Advisor of the Year**

National Independent Brokers Forum

Portfolio strategy for multi-property estates



2025

**ICEA Outstanding Advisor**

Institute for Chartered Estate Advisors

Off-market matchmaking & negotiation architecture

### Expertise



**Luxury estates**

Private previewsEstate-caliber staging



**Market analysis**

Hyper-local compsPricing scenarios



**Investment strategy**

Cap stacksRenovation ROI



Brand



Stats



Homes



Edge

### ### 7.1 How to capture

1. Run ``flutter run -d macos`` (or your device).
2. Navigate each tab; capture window (**`Cmd + Shift + 4`**, then **`Space`** on macOS).
3. Save using the names in ``docs/screenshots/README.md``.

### ### 7.2 Placeholder gallery (replace with your images)

> **Action required:** *Add your PNG files, then uncomment or update paths.*

#	Tab / topic	File to add
1	Brand — hero + mission	`docs/screenshots/01-brand-home.png`
2	Featured successes carousel	`docs/screenshots/02-featured-carousel.png`
3	Stats dashboard	`docs/screenshots/03-stats-tab.png`
4	Property list	`docs/screenshots/04-homes-list.png`
5	Property detail (Hero)	`docs/screenshots/05-property-detail.png`
6	Edge — awards	`docs/screenshots/06-edge-awards.png`
7	Edge — expertise chips	`docs/screenshots/07-edge-expertise.png`
8	Firebase Console	`portfolio/main`   `docs/screenshots/08-firebase-console.png`

Example markdown once files exist:

```
```markdown
```

```
![[Brand home]](docs/screenshots/01-brand-home.png)
![[Stats]](docs/screenshots/03-stats-tab.png)
![[Firestore]](docs/screenshots/08-firebase-console.png)
```

```
```
```

```

```

## ## 8. Architecture and development process

### ### 8.1 Layered architecture

```
```
```

```
main.dart
```

```
→ PortfolioRepository.initializeFirebase()
→ AgentPortfolioShell (StreamBuilder<PortfolioBundle>)
  → AgentHomeScreen / Stats / Homes / Edge
    → reusable widgets (glass, cards, images)
```

```
```
```

### ### 8.2 Development phases

1. **UI prototype** — Dark luxury theme, four tabs, static content.
2. **Modular refactor** — Split `screens/`, `widgets/`, `models/`, `data/`.

3. **Firestore integration** — ``PortfolioBundle``, ``PortfolioRepository``, Firestore document ``portfolio/main``.
4. **Live stream** — ``watchPortfolio()`` rebuilds UI on Firestore changes.
5. **Seed workflow** — In-app seed button + local fallback for tests.
6. **Documentation** — README, Firestore rules, screenshot folder.

### ### 8.3 Assignment requirements mapping

| Requirement                                                   | Implementation                                                                        |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Dark Luxury theme                                             | <code>`luxury_dark_theme.dart`</code>                                                 |
| Personal branding                                             | Brand + Edge tabs; mission + awards                                                   |
| <code>`CustomScrollView`</code> + <code>`SliverAppBar`</code> | <code>`agent_home_screen.dart`</code>                                                 |
| Featured successes (horizontal)                               | <code>`featured_success_carousel.dart`</code>                                         |
| Stats grid (3 cards)                                          | <code>`stats_dashboard_screen.dart`</code>                                            |
| Property card + Hero + badge                                  | <code>`luxury_property_card.dart`</code> + <code>`property_detail_screen.dart`</code> |
| Expertise + chips                                             | <code>`expertise_screen.dart`</code>                                                  |
| Glassmorphism                                                 | <code>`glass_surface.dart`</code> , <code>`glass_bottom_navigation.dart`</code>       |
| Unsplash images                                               | URLs in seed / Firestore                                                              |
| <b>Firestore storage</b>                                      | <code>`portfolio_repository.dart`</code> + Firestore                                  |
| <b>README</b>                                                 | This file                                                                             |

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### ## 9. Features by screen

| Tab          | Screen                                | Data from Firestore                                                                                                                |
|--------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Brand</b> | <code>`AgentHomeScreen`</code>        | <code>`agentName`</code> , <code>`profileHeroImageUrl`</code> , <code>`missionStatement`</code> , <code>`featuredSuccesses`</code> |
| <b>Stats</b> | <code>`StatsDashboardScreen`</code>   | <code>`stats[]`</code>                                                                                                             |
| <b>Homes</b> | <code>`PropertyShowcaseScreen`</code> | <code>`listings[]`</code>                                                                                                          |
| <b>Edge</b>  | <code>`ExpertiseScreen`</code>        | <code>`awards[]`</code> , <code>`expertisItems[]`</code>                                                                           |

---

### ## 10. Testing

```
```bash
```

flutter analyze

flutter test

...

| Test file | Purpose |

|-----|-----|

| `test/local\_portfolio\_seed\_test.dart` | Seed completeness + Firestore map round-trip |

| `test/widget\_test.dart` | Home + Stats tab smoke tests (local fallback, no Firebase) |

Tests do ****not**** require a live Firebase project.

11. Troubleshooting

| Issue | Solution |

|-----|-----|

| `kernel\_snapshot\_program failed` | Run `flutter analyze`; fix Dart errors (often wrong `import` paths). |

| App shows local data only | Run `flutterfire configure`; ensure `projectId` is not `REPLACE\_ME`. |

| Seed button never appears | Firebase must initialize; check console logs in debug. |

| Images blank | Need internet for Unsplash; check macOS network entitlement. |

| `hardware\_keyboard.dart` assertion (macOS debug) | Harmless Flutter desktop bug; full restart or `flutter run --release`. |

| Firestore permission denied | Deploy `firestore.rules` or use test mode in console. |

References

- [Flutter documentation](<https://docs.flutter.dev/>)

- [FlutterFire setup](<https://firebase.google.com/docs/flutter/setup>)

- [Cloud Firestore](<https://firebase.google.com/docs/firestore>)

- [Unsplash](<https://unsplash.com>)